CHAPTER 3

CHEMICAL KINETICS

CHAPTER ESSENCE AT A GLANCE:
Chemical reaction rates, rate laws, order and molecularity, integrated rate equations, half-life, and temperature dependence via Arrhenius equation.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Reaction Rate & Differential Expressions:** Rate quantifies the speed of reactant consumption or product formation per unit time; average rate measures concentration change over a finite interval, whereas instantaneous rate represents the derivative slope of the concentration-time curve.

• **Order vs Molecularity:** Order is an experimentally deduced sum of reactant concentration exponents in the rate law that can be zero, fractional, or negative; molecularity is a theoretical count of reacting species colliding simultaneously in an elementary step and can only take small positive integer values.

• **Integrated Rate Laws & Half-Life Kinetics:** Zero-order reactions proceed at a constant speed with a half-life directly proportional to initial concentration; first-order reactions exhibit an exponential concentration decay with a constant half-life that is completely independent of initial reactant concentration.

• **Arrhenius Temperature Dependence & Activation Energy:** Reaction rates rise exponentially with temperature because a higher thermal fraction of colliding molecules overcomes the activation energy threshold barrier.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$\text{Rate} = -\frac{1}{a}\frac{d[\text{A}]}{dt} = +\frac{1}{b}\frac{d[\text{B}]}{dt} = k[\text{A}]^x[\text{B}]^y$$

Rate of reaction and differential rate law (Overall order = x + y)

$$k = \frac{[\text{R}]_0 - [\text{R}]}{t} \quad \text{and} \quad t_{1/2} = \frac{[\text{R}]_0}{2k}$$

Integrated rate equation and half-life for zero-order reaction

$$k = \frac{2.303}{t}\log\left(\frac{[\text{R}]_0}{[\text{R}]}\right) \quad \text{and} \quad t_{1/2} = \frac{0.693}{k}$$

Integrated rate equation and half-life for first-order reaction

$$\log\left(\frac{k_2}{k_1}\right) = \frac{E_a}{2.303R}\left(\frac{T_2 - T_1}{T_1T_2}\right)$$

Arrhenius equation for temperature dependence and activation energy E a

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Order is experimental (can be 0, fraction, negative). Molecularity is theoretical whole number (1, 2, 3), never zero or fractional.

• **Alert:** For a first order reaction, doubling the initial concentration has ZERO effect on half-life!

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Define rate of a chemical reaction. Differentiate between average rate and instantaneous rate. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Rate]** Change in concentration of reactant or product per unit time: Rate = -d[R] divided by dt = +d[P] divided by dt (mol L^{-1} s^{-1}). [1 Mark]

• **[Step 2: Average vs Instantaneous]** Average rate r avg = -Δ[R] divided by Δt over finite time interval Δt. Instantaneous rate r inst = -d[R] divided by dt at a specific moment (slope of tangent). [1 Mark]

• **Key Exam Result:** r avg = -Δ[R] divided by Δt vs r inst = -d[R] divided by dt

2. Differentiate between order of a reaction and molecularity of a reaction (four points). [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Order vs Molecularity]** 1. Order is sum of powers of concentration terms in rate law (experimental); Molecularity is number of reacting species colliding simultaneously in elementary reaction (theoretical). 2. Order can be zero, fractional, or integer; Molecularity can only be whole numbers (1, 2, 3), never zero or fractional. 3. Order applies to elementary and complex reactions; Molecularity has meaning only for elementary steps. [3 Marks]

• **Key Exam Result:** Order (experimental, fractional/zero) vs Molecularity (theoretical, whole numbers)

3. Derive the integrated rate equation and half-life expression for a zero order reaction. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Integrated Rate Equation]** Rate = -d[R] divided by dt = k $[R]^0$ = k implies d[R] = -k dt. Integrating: [R] = -kt + $[R]_0$ implies k = ($[R]_0$ - [R]) divided by t. [2 Marks]

• **[Step 2: Half-Life]** At t = t_1 divided by $_2$, [R] = $[R]_0$ divided by 2. k = ($[R]_0$ - $[R]_0$ divided by 2) divided by t_1 divided by $_2$ = $[R]_0$ divided by ($2\ t_1$ divided by $_2$) implies t_1 divided by $_2$ = $[R]_0$ divided by (2k). Directly proportional to initial concentration. [2 Marks]

• **Key Exam Result:** k = ($[R]_0$ - [R]) divided by t | t_1 divided by $_2$ = $[R]_0$ divided by (2k)

4. Derive the integrated rate equation and half-life expression for a first order reaction. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Integrated Equation]** Rate = -d[R] divided by dt = k [R] implies d[R] divided by [R] = -k dt. Integrating: ln([R] divided by $[R]_0$) = -kt implies k = (2.303 divided by t) $\log_{10}$($[R]_0$ divided by [R]). [2.5 Marks]

• **[Step 2: Half-Life]** At t_1 divided by $_2$, [R] = $[R]_0$ divided by 2: k = (2.303 divided by t_1 divided by $_2$) $\log_{10}(2)$ = (2.303 × 0.3010) divided by t_1 divided by $_2$ = 0.693 divided by t_1 divided by $_2$ implies t_1 divided by $_2$ = 0.693 divided by k. Independent of $[R]_0$! [1.5 Marks]

• **Key Exam Result:** k = (2.303 divided by t) log($[R]_0$ divided by [R]) | t_1 divided by $_2$ = 0.693 divided by k

5. Show that for a first order reaction, time required for 99.9% completion is 10 times the half-life. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Pseudo First Order]** Reaction which is truly of higher order but follows first-order kinetics due to one reactant being in huge excess. [1 Mark]

• **[Step 2: Example]** Acid-catalyzed hydrolysis of ethyl acetate: $CH_3COOC_2H_5 + H_2O$ (excess) --> $[H^+]$ $CH_3COOH + C_2H_5OH$. Rate = k' $[CH_3COOC_2H_5]$ $[H_2O]$. Since $[H_2O]$ is constant: Rate = k $[CH_3COOC_2H_5]$, where k = k' $[H_2O]$. [2 Marks]

• **Key Exam Result:** Rate = k [Ester]; water in large excess behaves as constant

6. What is a pseudo first order reaction? Give one example with chemical equation. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Arrhenius Equation]** k = A e^(-E a divided by RT), where A is frequency factor, E a is activation energy, R is gas constant, T is temperature. [1.5 Marks]

• **[Step 2: Logarithmic Form]** $\log_{10}$(k_2 divided by k_1) = (E a divided by 2.303 R) [(T_2 - T_1) divided by (T_1 T_2)]. [1.5 Marks]

• **Key Exam Result:** log(k_2 divided by k_1) = (E a divided by 2.303 R) [(T_2 - T_1) divided by (T_1 T_2)]

7. Write the Arrhenius equation. Explain the terms: frequency factor (A) and activation energy (E a). [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Definition of E a]** Minimum extra energy required by colliding reactant molecules to form activated complex or reach threshold energy (E a = E threshold - E reactants). [1 Mark]

• **[Step 2: Catalyst Effect]** A catalyst provides an alternative reaction pathway having a lower activation energy (E a' < E a). Consequently, a much greater fraction of molecules possess sufficient energy to react, dramatically increasing reaction rate. [2 Marks]

• **Key Exam Result:** Catalyst lowers activation energy E a, increasing reaction rate